“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 15

Any Reminders? Any Questions?

- Quiz on Friday on triple integrals in cartesian, cylindrical, and spherical coordinates.
- I will be away Wednesday-Thursday at a conference so no in-person class on 3/12 – watch videos instead!

ALVARO: Start the recording!

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Exercises in Double and Triple Integrals

Example:

T F The area of the semi-circle centered at the origin with radius R, above the x-axis, can be computed with the following double integral: $\int_{-\pi/2}^{\pi/2} \int_0^R r \, dr \, d\theta$.

Example:

T F The following iterated integrals describe the volume of the same region E.

$$\int_{-1}^{1}\int_{x^2}^{1}\int_{0}^{1-y} 1\, dz\, dy\, dx \qquad \int_{-1}^{1}\int_{0}^{1-x^2}\int_{x^2}^{1-z} 1\, dy\, dz\, dx$$

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Example:

T F The intersection of the surfaces given by $\phi = \dfrac{\pi}{4}$ and $\rho = 2$ is a circle of radius 2.

Example:

Let E be the region above the xy-plane, below $z = \sqrt{x^2 + y^2}$, and inside $x^2 + y^2 = 9$. Write a triple integral below that represents the volume of E in two of the three coordinate systems that we have studied. For each part, circle or write the name of the coordinate system that you are using.

(a) Cartesian / Cylindrical / Spherical

Example:

Let E be the region above the xy-plane, below $z = \sqrt{x^2 + y^2}$, and inside $x^2 + y^2 = 9$. Write a triple integral below that represents the volume of E in two of the three coordinate systems that we have studied. For each part, circle or write the name of the coordinate system that you are using.

(a) Cartesian / Cylindrical / Spherical

Example:

Let D be the region in the xy-plane bounded by $y = x^2$ and $x = \frac{1}{8}y^2$.

(a) Sketch the region D. Clearly indicate the region by shading and make sure to label the equations of any boundary curves.

Example:

(b) Write a double integral that represents the area of the region D in the order $dy\,dx$.

Example:

(c) Write a double integral that represents the area of the region D in the order $dx\,dy$.

Example:

(a) Rewrite the following iterated integral using polar coordinates:

$$\int_{-2}^{2}\int_{0}^{\sqrt{4-x^2}} (x^2+y^2)^{3/2}\, dy\, dx.$$

(b) Evaluate the integral you wrote in part (a).

Example:

(a) Rewrite the following iterated integral using polar coordinates:

$$\int_{-2}^{2}\int_{0}^{\sqrt{4-x^2}} (x^2+y^2)^{3/2}\,dy\,dx.$$

(b) Evaluate the integral you wrote in part (a).

Example:

Set up the triple integral $\iiint_E x^2 e^y \, dV$, where E is the region enclosed by the surfaces $z = 9 - x^2$ and $z = 9 - y$ in the first octant. **Do not evaluate.**

Example:

Set up the following integral using either cylindrical or spherical coordinates. **Do not evaluate.**

The volume of the region E, where E lies above the xy-plane, below the cone $\phi = \pi/6$ and inside of the sphere $x^2 + y^2 + z^2 = 4$.

Example:

Questions?

Thank you

Until next time.